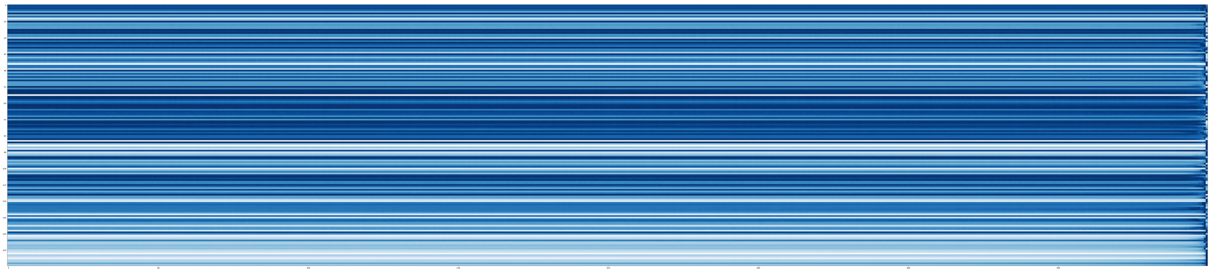
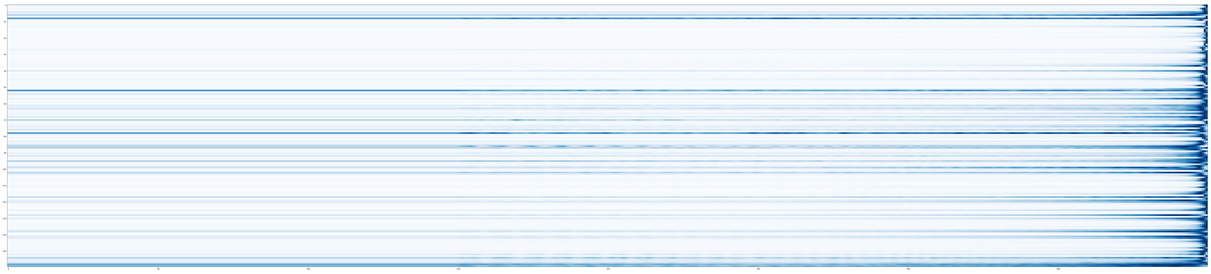
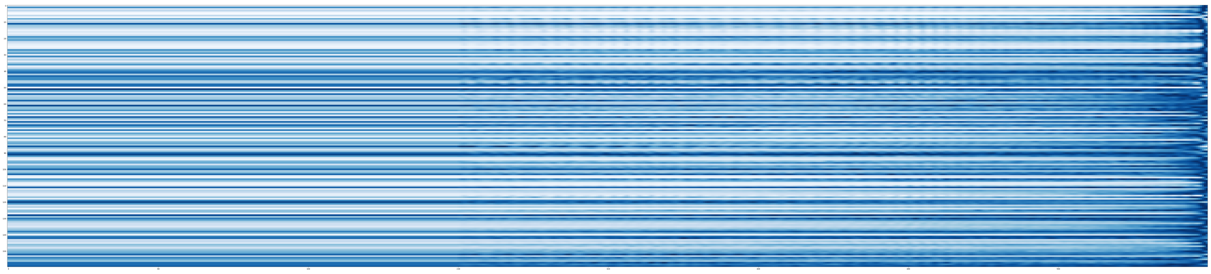


(a) Term (a).



(b) Term (b).



(c) Term (d).

Figure 7: Visualization of the three terms in computing the attention score. Each row corresponds to a attention head and each column corresponds to a relative location.

E Generated Text

In this section, we present some generated text from our best model trained the Wikitext-103 dataset. We seed the our Transformer-XL with a context of at most 512 consecutive tokens randomly sampled from the test set of Wikitext-103. Then, we run Transformer-XL to generate a *pre-defined* number of tokens (500 or 1,000 in our case). For each generation step, we first find the top-40 probabilities of the next-step distribution and sample from top-40 tokens based on the re-normalized distribution. To help reading, we detokenize the context, the generated text and the reference text. Three generated examples are shown in Tables 11, 12, and 13. Note that we do not perform any cherry picking and present the first three examples we generate in the paper. In the text, "= text =", "= = text = =" and "= = = text = = =" denote the Wikipedia page tile, section title and subsection title, respectively, due to the original data preprocessing procedure of Wikitext-103 (Merity et al., 2016).

As we can see, though only trained on 100M tokens, Transformer-XL is a strong model at generating long text articles, particularly in the following aspects:

- Transformer-XL is able to structurally maintain the sectional arrangement of Wikipedia.
- Transformer-XL manages to semantically stay on the same topic throughout the course of generation.
- Long-range references are common in the generated text.
- Transformer-XL often generates novel content that is not present in the training data.

For more detailed explanation of the interesting observations in each example, please refer to the corresponding caption.